Kopibara RLab Agent: An autonomous ML researcher that searches over executable hypotheses, one measured code change at a time.

Phanuphat Srisukhawasu
phanuphat.srisukhawasu@gmail.com
Kopibara Team

Supachod Trakansirorut
spchdt@gmail.com
Kopibara Team

September 1, 2026

1 Objective and evaluation

The task is to automate the engineering loop for a recommendation ranker: inspect the benchmark contract, construct features, train and tune a model, evaluate on public validation data, and revise the code. KuaiRand ranks logged impressions within each user. The relevance label is long-view feedback, and the primary score is the mean of GAUC and nDCG@5. KuaiRand-Pure is the required benchmark; KuaiRand-1K is the larger bonus variant.

The controller develops against train and validation splits only. Test rows are used only to produce the final score file, after their labels and feedback have been masked. The hidden split is never read during search. The organizer's convergence rule is an improvement of more than 0.002 over three consecutive iterations, with 50 iterations and six hours as backstops.

The official Pure Factorization Machine reference is 0.6016 on validation primary. For 1K, the starter kit does not publish an official reference, so the run measures the supplied FM pipeline at 0.6079 and labels it as a runtime reference. This distinction matters: the Pure comparison is against the organizer's published number, while the 1K comparison is only an internal benchmark.

2 Agent design

The controller uses GPT-5.6 Luna with low reasoning effort as its planner. Its design follows the executable code-search framing of AIDE and the reproducible benchmark discipline represented by MLE-bench. At each turn, the planner receives the benchmark contract, the current solution tree, the best measured node, and the source of the node selected as parent. It returns one testable hypothesis and a small set of exact-match replacements.

The runner applies a replacement only when its anchor occurs exactly once. The planner may edit the ranking source only, with at most four replacements, and the candidate must retain the existing command-line interface and validation output. A denylist rejects new network, shell, dynamic-import, or code-evaluation paths. Candidate processes run without the planner's credentials, so generated code cannot call the language model.

Each candidate is compiled and run in a bounded subprocess. The runner parses the candidate's validation metrics, uses the saved metric artifact when stdout is incomplete, and stores the hypothesis, exact diff, metrics, token counts, command, and recovery events. The controller keeps the highest-scoring measured node as the next parent. A failed candidate gets one repair request; a second failure is recorded as a recovered failure and cannot become a parent.

3 Seed pipeline

The seed is a grouped LightGBM ranker. It combines categorical context fields for the user, video, author, tab, duration bucket, hour, random-exposure flag, date, 15-minute time bucket, weekday, and released extra fields with continuous duration and time features. The ranking groups are users, matching the evaluation unit rather than treating the dataset as an ungrouped binary-classification problem.

The main inductive bias is a chronological history feature builder. For each user, video, author, and user-video pair, it records prior interaction count, feedback rates, log-transformed cumulative feedback, the last-seen timestamp transform, and the log time since the previous observation. All released feedback signals are available to these histories as auxiliary signals, while long-view feedback remains the only scored label. The state is updated only after an observed train or validation row, so a row cannot use its own outcome or a later outcome. Test rows receive features but do not update the state.

4 Search results

The selected validation results are summarized in Table 1. The 1K primary rises from 0.6079 for the measured FM reference to 0.7498. The Pure primary rises from the official validation reference of 0.6016 to 0.6299.

Benchmark	Reference	GAUC	nDCG@5	Primary	Delta
KuaiRand-Pure	0.6016 official	0.7059	0.5538	**0.6299**	**+0.0283**
KuaiRand-1K	0.6079 measured	0.7107	0.7888	**0.7498**	**+0.1419**

Table 1: Best validation results. The 1K reference is the measured FM pipeline supplied with the starter kit.

The Pure seed was already the best measured node at 0.6299. The search then checked three children: a planned XE-NDCG change that was not effective because the fixed Pure command continued to use LambdaRank with truncation 10 and disabled query normalization. Their primary scores were 0.6299, 0.6264, and 0.6268 respectively. The unchanged branch therefore does not test XE-NDCG; it exposes a command-wiring limitation in the search space.

The 1K run found a much larger improvement. The seed scored 0.6038, below the measured FM reference. Enabling grouped rank-XE-NDCG raised primary to 0.6920, with nDCG@5 increasing to 0.6971. Increasing the number of leaves from 31 to 63 then raised GAUC to 0.7034 and nDCG@5 to 0.7588, for primary 0.7311. A latest-feedback feature extension, smaller leaves, path smoothing, and truncation levels 10 and 3 were all tested from that parent and rejected.

The next accepted change added L2 regularization of 2.0, moving primary to 0.7487. The planner then tested stronger, weaker, and intermediate L2, more boosting rounds, a slower learning rate, L1, feature subsampling, and categorical smoothing. Every one was lower than the L2=2.0 parent or unchanged. Finally, a histogram resolution of 511 improved primary to 0.7493. A resolution of 1023 fell to 0.7473, 640 reached 0.7475, and the intermediate resolution of 767 became the best node at 0.7498.

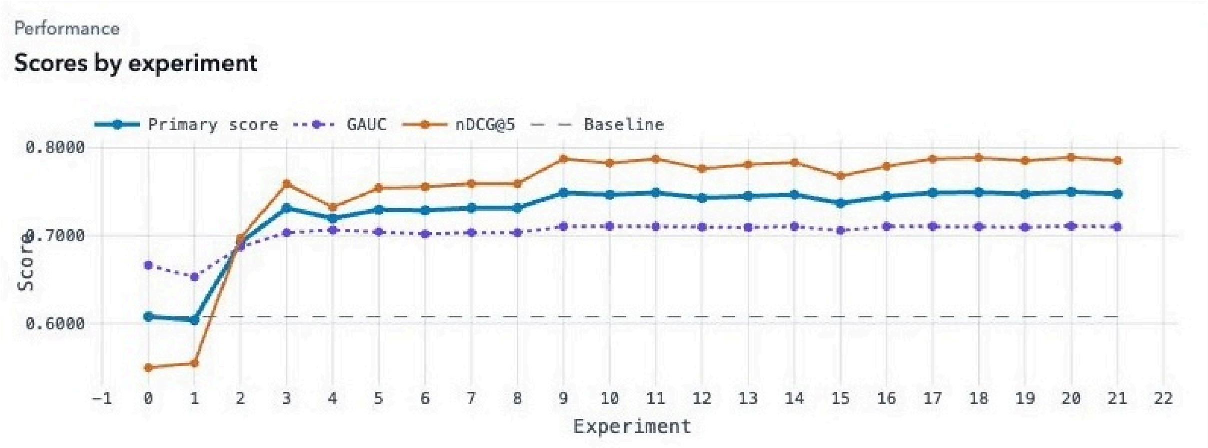

Figure 1: 1K validation trajectory. The lines show primary, GAUC, nDCG@5, and the measured reference.

The trajectory in Figure 1 shows two different regimes. The objective change and tree capacity produced the large gains; once the ranker reached 0.7311, the search became local regularization and quantization tuning. After 0.7311, the successful changes were smaller and more local: L2 regularization and histogram resolution. The controller kept only improvements and preserved the best checkpoint when later trials failed or regressed.

5 Complete decision record

Table 2 records the primary score used for parent selection. “Rejected” means the candidate ran but did not exceed the current best. “Failed” means both the original candidate and its repair timed out.

Iter.	Decision tested	Primary	Outcome
0	Seed: context plus four lagged multi-feedback histories	0.6038	Kept
1	Use grouped rank-XE-NDCG instead of LambdaRank	0.6920	Kept
2	Increase the number of leaves from 31 to 63	0.7311	Kept
3	Add latest feedback at each history scope	0.7192	Rejected
4	Reduce minimum data in each leaf to 20	0.7292	Rejected
5	Add path smoothing of 20.0	0.7285	Rejected
6	Increase ranking truncation to 10	0.7311	Rejected
7	Reduce ranking truncation to 3	0.7311	Rejected
8	Add L2 regularization of 2.0	0.7487	Kept
9	Increase L2 regularization to 4.0	0.7465	Rejected
10	Allow 600 boosting rounds	0.7487	Rejected
11	Reduce L2 regularization to 1.0	0.7427	Rejected
12	Use learning rate 0.03 and 700 rounds	0.7448	Rejected
13	Use intermediate L2 value 2.5	0.7466	Rejected
14	Add mild L1 regularization	0.7366	Rejected
15	Use feature fraction 0.85	0.7445	Rejected
16	Smooth categorical splits with categorical smoothing of 20.0	0.7487	Rejected
17	Increase histogram resolution to 511	0.7493	Kept
18	Increase histogram resolution further to 1023	0.7473	Rejected

Iter.	Decision tested	Primary	Outcome
19	Use intermediate histogram resolution of 767	0.7498	Kept
20	Tune near the optimum with histogram resolution of 640	0.7475	Rejected
21	Reduce XE-NDCG sigmoid scale to 0.5	–	Failed: timeout; retained 019

Table 2: Complete 1K search sequence recorded by the autonomous controller.

The Pure sequence reached the organizer convergence rule after three children. The first child proposed XE-NDCG, but the fixed command overrode the candidate's new default and continued to use LambdaRank. That score is therefore a wiring diagnostic, not evidence about XE-NDCG itself.

Iter.	Decision tested	Primary	Outcome
0	Seed: grouped LightGBM with leakage-safe histories	0.6299	Kept
1	XE-NDCG support planned; command still used LambdaRank	0.6299	No-op; rejected
2	Increase ranking truncation to 10	0.6264	Rejected
3	Disable query-size lambda normalization	0.6268	Rejected

Table 3: Pure search sequence. Scores are validation primary.

6 Reproducibility, safety, and accounting

Each trial is reproducible from its hypothesis, exact code diff, validation metrics, subprocess command, token count, and recovery event. The generated outputs contain 170,588 row-aligned Pure records and 4,132,081 row-aligned 1K records. Neither run required manual intervention or used hidden-test labels.

Benchmark	Run ID	Iters.	Wall-clock	LLM tokens	Manual	Stop
KuaiRand-Pure	20260831T111545Z	3	5m 5s	12,901	0	Converged
KuaiRand-1K	20260831T124656Z	21	6h 7m	115,212	0	Wall-clock cap

Table 4: Resource and autonomy accounting for the two runs.

One planner response exceeded the four-edit limit and was retried. At iteration 21, the candidate and its repair both timed out after 375.92 seconds, so the controller retained the best node from iteration 19 at 0.7498. The failure could not replace the selected checkpoint.

All scores above come from the fixed training and validation split. The Pure comparison uses the official reference; the 1K comparison uses the measured FM reference described in Section 1. The selected checkpoints produce row-aligned evaluation outputs for both variants.